Project Overview

Incremental Data-Driven Robust Crew Pairing Optimization

Under Evolving Uncertainty

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Problem

- Crew schedules must cover all flights.
- Delays can break crew connections.
- Rare disruptions are hard to model.

Main idea

- Use quantile-based delay bounds.
- Model harmful one-sided delays.
- Balance robustness and cost.

Core message: build crew schedules that can survive realistic flight delays.

Solution Framework

Historical delay data → Quantile uncertainty set → Select crew pairings → Recover schedule

Two-stage robust optimization

- First stage: choose feasible crew pairings before disruption is known.
- Second stage: choose wait, swap, or cancel actions after delays appear.
- Nested C&CG and column generation add only important scenarios and pairings.

Conclusion

Main contribution: incremental re-optimization

- New data update the uncertainty bounds.
- If uncertainty shrinks, the old robust plan remains feasible.
- If uncertainty expands, add new cuts and re-optimize.
- The model avoids rebuilding everything from zero.

Takeaway

A robust crew pairing method that learns from delay data and adapts as uncertainty evolves.